SOIL EROSION AND CONSERVATION · NOTE ST03

Spillway Components — Inlets, Conduits, Anti-Seep Collars, Outlets and Protection Works

Course topics: SWCE-503 T1, SWCE-503 T8

Parts of a structure

- Basic components: check dam, embankment and/or retaining wall, and spillway. A spillway is an **inlet**, a **conduit** and an **outlet**, and spillways are classified by the form of each.
- The **embankment** directs flow into the spillway and runs from it to high ground or a vegetated spillway; in earth dams it impounds water while the spillway passes excess storm water.

Inlet

- The upstream end through which water enters — a **box**, a **weir in a wall**, or a **culvert-type entrance**.
- **Box inlet:** a rectangular box open at the top and downstream end, with a baffle wall downstream; its front and two sides act as weirs, so flow enters at the upstream end and sides and leaves at the downstream end. Dikes or a headwall can direct the approach flow. Preferred for large flows through a narrow structure at low head.
- **Culvert-type entrance:** works as an orifice — round, square or rectangular, with square-edged, hood or flared entrance.
- Inlet geometry may be straight, box or curved; a curved inlet gains arch strength in masonry.

Free-flow capacity: $Q = C \cdot L \cdot H^{(3/2)}$

Average conditions: $Q = 1.77 \cdot L \cdot H^{(3/2)}$ (Q in m^3/s; L, H in m; accuracy $\pm 20\%$)

Q in litres/s, L and H in cm: $Q = 0.0177 \cdot L \cdot H^{(3/2)}$

L = crest length — the straight weir crest, three sides of a box inlet, or the circumference of an arch inlet; H = depth over the crest. Add 15–30 cm freeboard above the computed H. (Check: 1.77 × 1000/(100 × 100^1.5) = 0.0177.)

Inlet part	Function
Headwall	Masonry or concrete retaining wall at the upstream end, with enough base width and weight to resist overturning and other forces
Cutoff wall	Vertical wall below the inlet extending into the foundation to stop seepage under the structure
Headwall extension	Lateral extension of the inlet wall: prevents seepage, erosion and piping, guards against burrowing animals, reduces uplift and resists sliding
Wing walls	Join the abutments to the dike or banks and improve flow — upstream wings to the upstream banks, downstream wings to the downstream channel
Side walls	Hold stable fills and protect the structure from erosion
Toe wall	Shallow wall below the bed or floor that prevents piping and undermining and can support sloped pitching or the embankment face
Guardrails and trash racks	Protect people and animals and prevent clogging; a rack covers the inlet without disturbing flow, its vertical opening about twice its width, in reinforced concrete, steel or aluminium pipe

Conduit

- Receives water from the inlet and carries it through the structure in a definite channel — closed (box or pipe) or open (rectangular channel). Appurtenances: cutoff walls and anti-seep collars.
- **Anti-seep collars** around the conduit minimise seepage along it, lengthen the creep path and bond it to the embankment; their size depends on the head and the density of the embankment material.
- **Expansion joints** allow expansion, contraction and settlement — one for every 40 m² of flat slab, spaced 7.5–15 m along the length; not needed for conduits shorter than 15 m or where settlement is insignificant.
- **Monolithic conduit or pipe drop spillway:** where the total fall in the conduit exceeds the expected settlement, spread the fall over the whole length.
- **Riprap:** a layer of non-erodible material (rock fragments, bricks, stones) on embankment slopes, streams and canals, thick enough to resist waves and currents.

Outlet

- The outlet discharges into the downstream channel at a safe velocity and dissipates (or converts to potential energy) the energy gained in the fall, protecting both the structure and the channel. Outlet wing walls guard against swirling flow; toe walls prevent the apron being undermined.
- **Downstream transition:** the channel section that changes gradually from the conduit to the downstream channel, measured along the flow. Protect it against the high outlet velocity with brick lining or riprap about 30 cm thick — or, where grass establishes quickly, a 6–10 m sodded strip.
- **Outlet design** must not reduce channel performance or stability, and must avoid scour and silt deposits that would interfere with spillway discharge.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. US Bureau of Reclamation, Design of Small Dams and Hydraulic Design of Stilling Basins and Energy Dissipators (EM 25). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The weir formula for free flow over a spillway inlet (average conditions, metric) is

(a) Q = 1.77LH^1.5 (b) Q = 0.0177LH^2.5
(c) Q = 3.33LH^1.5 (d) Q = 1.84LH

Q2. For a box inlet, crest length L equals

(a) one side (b) two sides
(c) three sides (d) four sides

Q3. A culvert-type entrance works on the

(a) weir principle (b) orifice principle
(c) siphon principle (d) jump principle

Q4. Anti-seep collars mainly

(a) support the pipe (b) increase creep length along the conduit
(c) trap trash (d) dissipate energy

Q5. Expansion joints are not needed when the conduit is shorter than

(a) 5 m (b) 15 m
(c) 40 m (d) 100 m

Q6. Riprap protecting a downstream transition is about

(a) 5 cm (b) 30 cm
(c) 1 m (d) 2 m

Q7. Freeboard above the computed head over an inlet is

(a) 5 cm (b) 15–30 cm
(c) 1 m (d) 2 m

Answers

Q1 — (a) Q = 1.77LH^1.5
Q2 — (c) three sides
Q3 — (b) orifice principle
Q4 — (b) increase creep length along the conduit
Q5 — (b) 15 m
Q6 — (b) 30 cm
Q7 — (b) 15–30 cm